CS341 · COMPUTER GRAPHICS

Midterm Review

The whole midterm in one file

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What's inside

01 What comes from each chapter

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06 Bresenham: (3,3) to (10,7)

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01 What comes from each chapter

Chapter	Out of 20	Question type
Chapter 1 (displays and CRT)	**7 – 8**	Definitions, true/false, MCQ, CRT figure, aspect ratio
Line drawing (DDA / Bresenham)	**4 – 5**	One problem: table, points and drawing
Chapter 3 (colors)	**3 – 5**	Color mixing and definitions (blanks)
Chapter 2 (vectors)	**2 – 5**	True/false and definitions
Introduction	**2 – 3**	Definition of CG and applications

Distribution based on midterms 472 and 461.

02 Introduction

Question	Answer
Define CG	**Creation, storage, and manipulation of models and images**
Main tasks	Modeling · Rendering · Animation
NOT a main task	Data encryption
Applications	CAD · Medical · Entertainment → **All of the above**
"Used only for entertainment"	**False**
Goal	Synthetic images indistinguishable from reality

03 Chapter 1: displays

Term	Answer
Random scan	The beam draws lines directly · picture in the **display list**
Raster scan	Scans top to bottom, line by line · picture in the **frame buffer**
A CRT emits	**Electrons**
Persistence	Time for the light to decay to **one-tenth** of its intensity
Refresh rate	How many times per second the picture is redrawn
Resolution	Maximum number of points without overlap
Screen clarity	resolution · **dot pitch** · refresh rate
Input device	Mouse (not monitor, printer or speaker)
Hard-copy devices	Laser · Ink-jet · Film recorder · Electrostatic · Pen plotter

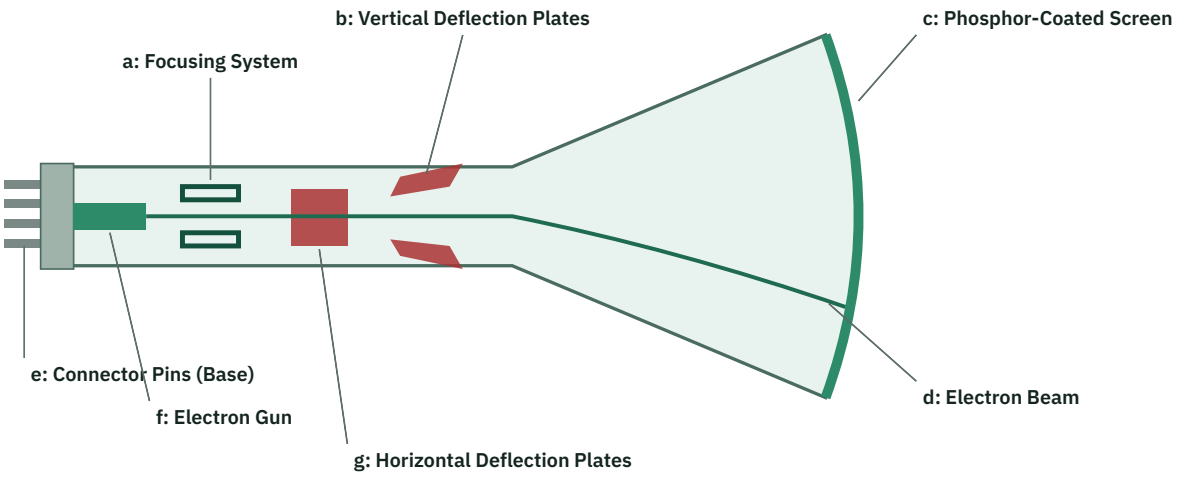
Aspect ratio

Why? To maintain geometric features: a square stays a square and a circle does not become an ellipse.

$$AR = \frac{\frac{\text{vertical points}}{\text{height}}}{\frac{\text{horizontal points}}{\text{width}}}$$

461: $AR = \frac{\frac{1080}{5}}{\frac{1920}{10}} = \frac{216}{192} = 1.125$

The CRT figure



From the left: pins → electron gun → focusing system → horizontal plates → vertical plates → screen (the beam hits the screen).

Absolute vs Relative

Absolute: the actual position, measured from the origin.

Relative: an offset from the last position (current position). (3, 8) + (2, −1) → (5, 7)

04 Chapter 2: vectors

Statement	Answer
A vector has both magnitude and direction.	**True**
Vector from P to Q	**Q – P**
Sum two points results in a valid point.	**False**
The dot product of two perpendicular vectors is zero.	**True**
A unit vector has a length greater than 1.	**False** (= 1)

05 Chapter 3: colors

Screen: RGB (adding light)

`R + G = Yellow`
`G + B = Cyan`
`R + B = Magenta`
`R + G + B = White`

Printer: CMY (inks)

`Y + M = Red`
`M + C = Blue`
`C + Y = Green`
`C + M + Y = Black`

Term	Meaning
Tint	Color + white
Shade	Color + black
Tone	Color + white + black
Color gamut	All colors produced by mixing the primary colors

06 Bresenham: (3,3) to (10,7)

1) $\Delta x = 10 - 3 = 7$ $\Delta y = 7 - 3 = 4$

2) $2\Delta y = 8$ $2\Delta y - 2\Delta x = -6$

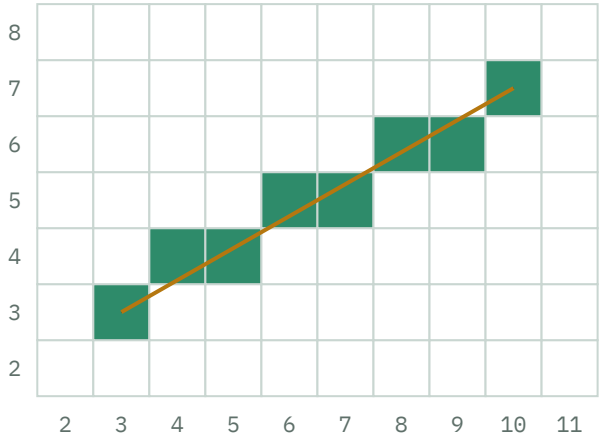
3) $p0 = 2\Delta y - \Delta x = 8 - 7 = 1$

4) x increases by 1 every step:

$p < 0 \rightarrow$ y stays $p = p + 8$

$p \geq 0 \rightarrow$ y + 1 $p = p - 6$

k	p_k	Decision	Next pixel	p_{k+1}
0	1	$p_k \geq 0$	(4, 4)	-5
1	-5	$p_k < 0$	(5, 4)	3
2	3	$p_k \geq 0$	(6, 5)	-3
3	-3	$p_k < 0$	(7, 5)	5
4	5	$p_k \geq 0$	(8, 6)	-1
5	-1	$p_k < 0$	(9, 6)	7
6	7	$p_k \geq 0$	(10, 7)	1



(3,3) (4,4) (5,4) (6,5) (7,5) (8,6) (9,6) (10,7)

07 DDA: (2,2) to (4,10)

1) dx = 2, dy = 8, steps = the larger = 8

2) xInc = $\frac{2}{8}$ = 0.25 yInc = $\frac{8}{8}$ = 1

3) round x every step (0.5 rounds up)

k	x	y	round(x)	round(y)
0	2	2	2	2
1	2.25	3	2	3
2	2.5	4	3	4
3	2.75	5	3	5
4	3	6	3	6
5	3.25	7	3	7
6	3.5	8	4	8
7	3.75	9	4	9
8	4	10	4	10

	DDA	Bresenham
Arithmetic	Floating point and rounding	Integers only
Speed and accuracy	Slower, less accurate	Faster, more accurate